

Integrated Risk Management in Civil Aviation: Contribution of Bayesian and AI Models to the Assessment of Cybersecurity Risks in Aviation Safety and Security

1st Wiyao BANAKINAO*

Higher Polytechnic School - LITA
Cheikh Anta Diop University of Dakar
Lomé, Togo
wiyao45@yahoo.fr

2nd Folly KOUÉVI

Higher Polytechnic School - LITA
Cheikh Anta Diop University of Dakar
Lomé, Togo
kouevi@yahoo.fr

3rd Mamadou BA

Higher Polytechnic School - LITA
Cheikh Anta Diop University of Dakar
Dakar, Senegal
ba.mamadou@esp.sn

4th KPANZOU Tchilabalo Abozou

University of Kara - LaMMASD
Kara, Togo
kpanzout@univkara.tg

5th Samuel OUYA

Cheikh Hamidou KANE Digital University (UN-CHK)
Dakar, Senegal
samuel.ouya@unchk.edu.sn

Abstract—The civil aviation sector is experiencing sustained growth, accompanied by a profound digital transformation of communication, navigation, surveillance and information-management systems. This increasing interconnection of critical infrastructures exposes aviation to new cybersecurity threats likely to affect safety, security and the continuity of operations. This paper proposes an integrated approach to the assessment and management of cyber risks applied to critical aeronautical systems, within a risk-based civil aviation security (AVSEC) logic. The methodology is based on the modelling of attack scenarios targeting the main technical domains and the overall set of critical infrastructures of international civil aviation, followed by the evaluation of three key components: the threat directed against the target (utility) denoted by (U), vulnerability (V) and consequences (C). Each component is quantified on three levels and weighted to compute a raw risk value. A Bayesian probabilistic model is then used to integrate uncertainty related to data and to the evolution of the threat context, while a non-linear interaction term accounts for dependencies between U , V and C . The approach is illustrated using representative scenarios, notably a denial-of-service attack on VHF communications. The results show that this assessment framework provides a relevant decision-support tool to strengthen the resilience of aeronautical systems, guide technical and organisational protection measures and concretely integrate cybersecurity into civil aviation security programmes, in line with the orientations of ICAO, EASA and risk-management guidelines [1], [2], [4]–[7], [12].

Index Terms—Cybersecurity, civil aviation, risk management, critical systems, Bayesian model, aviation security (AVSEC), aviation safety, critical infrastructures.

I. INTRODUCTION

Civil aviation remains one of the fastest-growing sectors worldwide, with an average annual increase of 4.2% in air

traffic before the COVID-19 pandemic and a rapid recovery now confirmed [1], [12]. To absorb this growth, aeronautical infrastructures have undergone a profound modernisation based on the integration of advanced digital technologies for communication (CPDLC, SATCOM), navigation (GNSS) and surveillance (ADS-B), which have become essential to ensure the performance and resilience of air traffic management (ATM) systems [2], [5]. However, this transition towards highly interconnected environments is accompanied by a significant increase in the exposure surface to cyberthreats, with a 131% increase in cyberattacks targeting the aviation sector between 2022 and 2023, and a nearly 600% rise in ransomware attacks between 2024 and 2025 [4], [12]. Critical infrastructures now exhibit vulnerabilities linked to the use of unsecured protocols, the absence of native encryption, dependence on GNSS signals that can be jammed or spoofed, the accessibility of radio communications via SDR, and weak or non-existent authentication mechanisms [5], [8].

Despite significant contributions in the scientific literature on vulnerabilities of NextGen systems [13], *e-enabled* architectures [14], and ADS-B, Mode S, VDL2 and GBAS communications [15]–[17], as well as on the protection of critical infrastructures [18], this work remains fragmented, essentially non-probabilistic, strongly oriented solely towards a safety-based perspective, and does not provide a complete integration of the AVSEC triad: threat directed at the target, vulnerabilities of the target and inherent consequences. This limits the ability to capture systemic phenomena and non-linear behaviours inherent in modern cyber-aviation environments.

The present paper proposes a unified methodological framework based on multi-risk probabilistic modelling that simultaneously integrates the dimensions of safety and security in aviation, relying on Bayesian models for quantitative risk assessment [6], [7], complemented by non-linear interaction terms and artificial intelligence mechanisms enabling dynamic updating [10], [11]. The objective is to provide a holistic, adaptable and operational tool aligned with the recommendations of ICAO, EASA and international risk-management guidelines [1], [2], [4], [5].

II. LITERATURE REVIEW

Over the last decade, numerous studies have addressed cybersecurity issues in communication, navigation and surveillance (CNS) systems, revealing critical vulnerabilities while developing risk-assessment methodologies. Sueki and Kim [13] provided one of the first overviews of vulnerabilities in the NextGen air traffic control system, highlighting weaknesses in the communication infrastructure, notably the absence of authentication and encryption in new data links, as well as the exposure of surveillance data to spoofing and jamming, although their analysis remains largely qualitative and does not provide a formal risk-assessment framework. Shaikh *et al.* [14] extended this analysis to *e-enabled* aircraft architectures, examining CPDLC and ACARS data links, GPS-based navigation and ADS-B surveillance, stressing the growing interconnection between on-board and ground networks, and identifying numerous vulnerabilities ranging from message injection and replay attacks to denial-of-service attacks on critical avionics buses.

Osechas *et al.* [15] focused on vulnerabilities of CNS infrastructures facing targeted radiofrequency interference, demonstrating the practical feasibility of jamming and spoofing VDL2, GBAS and secondary radar (SSR) signals using relatively low-cost equipment, thus characterising weaknesses at the signal level without embedding them in a broader risk-management process. Strohmeier and his collaborators produced a substantial body of work on the security of wireless aeronautical communications, empirically assessing threats against systems such as ADS-B and Mode S [16], comparing the theoretical attack surface with observed incidents and showing that, despite growing awareness, operational deployments still rely heavily on unauthenticated and unencrypted transmissions. In subsequent work on securing the air-ground link [17], they discussed attacks targeting VHF voice, CPDLC, ACARS and surveillance links, reviewing various cryptographic, intrusion-detection and multi-sensor verification countermeasures, and demonstrating that isolated technical solutions are insufficient to mitigate systemic risks linked to interdependencies between CNS components.

Gritzalis *et al.* [18] addressed cybersecurity and cyber-resilience in aviation from the perspective of critical infrastructure protection, discussing general risk-assessment principles and resilience metrics applicable to ATM environments, although their approaches remain relatively generic and not specific to the particularities of CNS/ATM and SWIM architectures. Elmarady and Rahouma [5] proposed a more integrated view

by developing a structured framework for cybersecurity risk assessment in CNS/ATM and SWIM environments, explicitly considering several system domains and modelling threats, vulnerabilities and impacts in a unified manner, enabling the prioritisation of mitigation measures according to their joint contribution to cybersecurity and flight-safety objectives.

Taken together, these works reveal several important gaps: most focus on specific subsets of CNS components without a holistic view, adopt either a qualitative or purely technical perspective without a systematic and reproducible risk-assessment methodology aligned with international aviation standards, and the articulation between cybersecurity and traditional safety risk management in ATM remains insufficiently explored, justifying the development of an integrated, probabilistic and evolving framework capable of simultaneously addressing safety and security dimensions in aviation.

III. RISK ASSESSMENT METHODOLOGY

The proposed methodology is based on international best practices in security and cybersecurity risk management from ICAO [2], NIST [6], EASA [4], IEEE [7], as well as on the cyber-risk assessment method for CNS and SWIM systems of Elmarady and Rahouma [5]. It combines structured modelling of risk factors, probabilistic classification and non-linear adjustment, complemented by artificial intelligence capabilities to dynamically adapt the level of criticality.

The approach is structured around three main components:

- 1) a **baseline risk score** derived from operational components;
- 2) a **probabilistic classification** using a Bayesian model;
- 3) a **non-linear and adaptive adjustment** that accounts for interactions and system evolution.

A. Modelling of Risk Components

The model retains three fundamental components, in line with risk approaches used in critical infrastructures [2], [6]:

- **Threat / Utility (U)**: attractiveness of the target to a malicious actor (terrorist or criminal), related to its strategic importance, operational role, symbolic value, redundancy and incident history.
- **Vulnerability (V)**: exposure of the target to threats, including security weaknesses (encryption, protocols, configuration, obsolescence) and safety weaknesses (location, physical measures, access control).
- **Consequences (C)**: impact of a compromise on flight safety, operational continuity and aviation security.

Each component is rated on a standardised scale [4]:

$$1 = \text{low}, \quad 2 = \text{moderate}, \quad 3 = \text{high}.$$

The selected weights are:

$$W_u = 0.4, \quad W_v = 0.5, \quad W_c = 0.1,$$

in order to reflect the central role of technical vulnerability in cybersecurity measures, while consequences may be mitigated by redundancy, fallback modes and ATM contingency plans.

These values remain adjustable depending on data or expert judgement.

B. Computation of the Baseline Risk Score

The continuous score combining the three components is defined by:

$$S = W_u U + W_v V + W_c C. \quad (1)$$

It constitutes the baseline for comparing scenarios. In the chosen configuration, vulnerability (V) influences S more strongly than U or C , reflecting its central role in cybersecurity programmes.

C. Probabilistic Classification Using a Bayesian Model

To classify each scenario into one of the usual categories (low, medium, high), the random variable L is introduced. The posterior probability is obtained using Bayes' rule:

$$P(L | U, V, C) = \frac{P(U, V, C | L) P(L)}{P(U, V, C)}. \quad (2)$$

Following NIST and IEEE approaches [6], [7], we adopt a conditional-independence assumption:

$$P(U, V, C | L) = P(U | L) P(V | L) P(C | L),$$

from which:

$$P(L | U, V, C) = \frac{P(U | L) P(V | L) P(C | L) P(L)}{P(U, V, C)}. \quad (3)$$

This assumption simplifies modelling, but is not always realistic: in CNS/ATM systems, the most critical elements benefit from strengthened protections (redundancies, fallback modes, contingency plans), which mechanically reduce their vulnerability.

D. Non-linear Adjustment via an Interaction Factor

To incorporate interactions between utility, vulnerability and consequences, a non-linear term is added to the model [7]:

$$f(U, V, C) = \alpha UV + \beta VC + \gamma UC,$$

where α , β and γ represent the intensity of these dependencies. For illustrative purposes, simulations use $\alpha = \beta = \gamma = 0.05$, producing moderate amplification when the three components are high. Larger coefficients would strengthen extreme cases, while smaller values would bring the model closer to an additive structure. More rigorous calibration would require sensitivity analysis or expert judgement.

E. Final Adjusted Risk Value

The final risk value is calculated as:

$$R_{\text{final}} = S + f(U, V, C), \quad (4)$$

providing a continuous indicator of criticality consistent with ICAO and EASA practices [2], [4]. For reasonable values of the interaction coefficients, simulations show that the ranking of scenarios remains generally stable, even though the numerical value of R_{final} varies. Robustness analyses can further refine this aspect.

F. Justification and Calibration of Coefficients

The weights (W_u, W_v, W_c) and coefficients (α, β, γ) are calibration parameters of the model. In this version, they were chosen to assign greater weight to vulnerability, while consequences play a smaller role due to redundancy and fallback mechanisms inherent in CNS/ATM systems. Their adjustment can be refined by several methods: expert analysis to reflect operational risk perception, and calibration on historical data to optimise the predictive performance of the model.

G. Integration of Artificial Intelligence

To improve the accuracy of risk estimation and adapt the model to the evolution of critical systems, an artificial-intelligence layer is added, in line with current trends in aviation safety and predictive cybersecurity [10], [11], and in continuity with work conducted on CNS and SWIM systems [5]. A machine-learning model f_θ automatically estimates the components (U, V, C) from an observation vector X aggregating information from event logs, SOC/SIEM alerts, technical configurations and the operational context.

We then have:

$$(U, V, C) = f_\theta(X) \quad (5)$$

Depending on the type and amount of available data, the model f_θ can be implemented using different machine-learning approaches: ensemble models (random forests, gradient boosting) for their robustness and partial interpretability, shallow neural networks adapted to tabular data, or autoencoders and anomaly-detection methods (Isolation Forest, One-Class SVM) to identify atypical behaviours in CNS/ATM data streams.

The AI-estimated risk score becomes:

$$S = W_u U + W_v V + W_c C,$$

and the final adjusted value is written:

$$R_{\text{final}}^{\text{AI}} = S + f(U, V, C).$$

An anomaly-detection module also provides a score $A(X)$ allowing the vulnerability to be dynamically modified as:

$$V' = V^{(1+\lambda A(X))},$$

in order to automatically adapt the risk level to the operational context. Training these models nevertheless requires labelled data or, failing that, synthetic datasets representing typical scenarios. Finally, AI integration must remain compatible with safety requirements: transparency, human oversight and rigorous model validation are essential in CNS/ATM environments.

IV. RESULTS AND DISCUSSION

A. Results

Simulations are performed by considering:

$$U, V, C \in \{1, 2, 3\},$$

and retaining the model weights and coefficients:

$$W_u = 0.4, \quad W_v = 0.5, \quad W_c = 0.1, \quad \alpha = \beta = \gamma = 0.05.$$

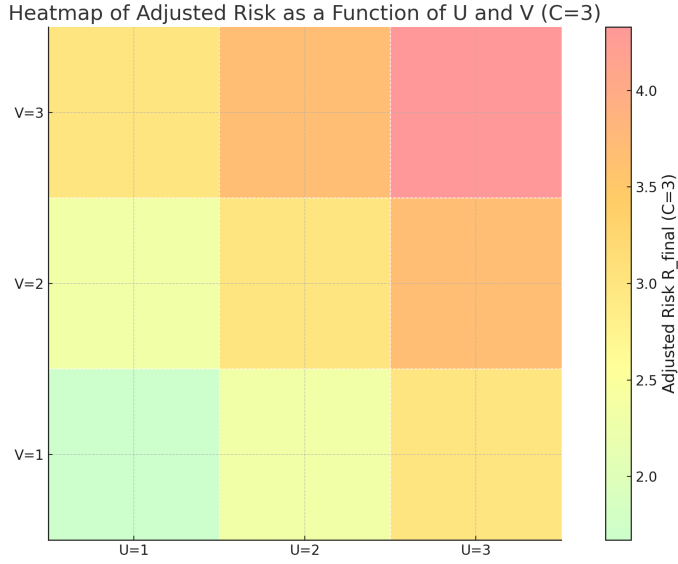


Fig. 1: Heatmap of adjusted risk R_{final} as a function of U and V for $C = 3$.

These values are purely illustrative and serve to analyse the behaviour of the model on a limited number of scenarios. They do not constitute definitive calibration: the objective is mainly to show how the structure of the model distinguishes risk levels and guides prioritisation. Calibration based on real data could lead to different values, or even modify the relative importance of U , V and C .

1) *Heatmap of Adjusted Risk R_{final}* : The first simulation consists in representing the surface of the adjusted risk R_{final} as a function of utility (U) and vulnerability (V), for a maximum consequence level ($C = 3$). Figure 1 shows the resulting heatmap.

This visual representation clearly shows that risk increases with U and V , with combinations ($U = 3, V = 3$) producing the highest values, consistent with criticality approaches applied to critical aeronautical infrastructures [2], [4], [5]. The non-linear term accentuates this effect when the components simultaneously reach their maximum values. Additional tests indicate that, for reasonable interaction coefficients, the general structure of the risk surface remains stable, which suggests good robustness of the model to parameter variations.

2) *Comparison of Representative Scenarios*: Three representative scenarios of cyberattacks frequently discussed in the aviation domain are evaluated: a DoS on VHF communications ($U = 1, V = 3, C = 3$), GNSS spoofing ($U = 1, V = 3, C = 2$) and a MITM attack on CPDLC ($U = 2, V = 2, C = 2$). Figure 2 illustrates the adjusted risk values R_{final} obtained for each of these scenarios.

The results show that the VHF DoS scenario presents the highest risk, followed by GNSS spoofing, which is also critical, while the MITM attack on CPDLC displays a moderate risk level.

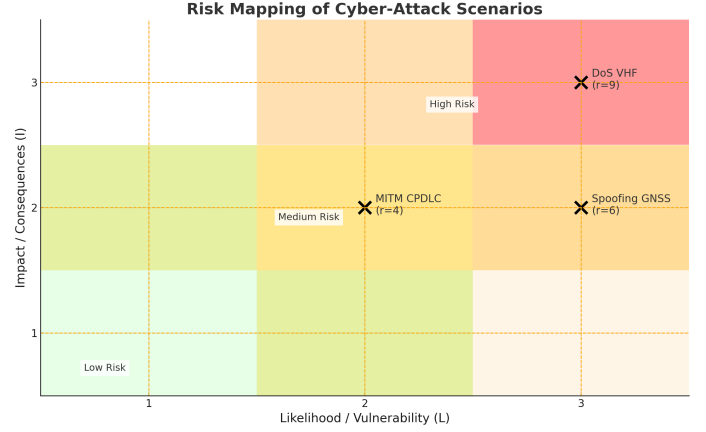


Fig. 2: Comparison of adjusted risk values R_{final} for three cyberattack scenarios.

B. Discussion

The proposed approach stands out from the state of the art through several methodological contributions that address the limitations identified in the existing literature. Unlike the essentially qualitative analyses of Sueki and Kim [13] and Shaikh *et al.* [14], which identify vulnerabilities in NextGen systems and *e-enabled* architectures without proposing quantitative metrics to compare scenarios or prioritise investments, our model introduces a continuous risk score R_{final} that enables objective and comparable evaluation of different threats. The results obtained for the three representative scenarios illustrate this quantitative differentiation capability, with respective values of 2.85 for VHF DoS, 2.55 for GNSS spoofing and 2.20 for CPDLC MITM, reflecting decreasing levels of criticality that are consistent with operational intuition but now measurable.

Whereas the works of Osechas *et al.* [15] and Strohmeier *et al.* [16], [17] convincingly demonstrate the technical feasibility of jamming and spoofing attacks on VDL2, GBAS, ADS-B and Mode S, they remain focused on experimental validation without integration into a structured cybersecurity risk-management process; our framework builds on these results by articulating three complementary dimensions—strategic utility (U), technical vulnerability (V) and operational consequences (C)—which make it possible to transform research results into prioritised operational recommendations.

The Bayesian model introduces a probabilistic dimension that is absent from traditional qualitative risk matrices [2], computing the posterior probability $P(L | U, V, C)$ for each risk level and thus quantifying the uncertainty associated with each classification, directly addressing the criticisms of Gritzalis *et al.* [18] regarding the static nature of conventional assessment methodologies. In contrast with the fragmentation observed in the literature, where analyses concentrate either on communications [13], [17], navigation [15] or surveillance [16], our approach adopts a systemic view of the complete CNS/ATM-SWIM architecture, and the non-linear interaction term $f(U, V, C) = \alpha UV + \beta VC + \gamma UC$ captures dependencies

between subsystems, modelling the fact that a simultaneous combination of high factors generates a disproportionate level of risk compared with the sum of individual components.

The operationalisation of the AVSEC triad threat–vulnerability–consequence represents a significant advance over existing work that treats the *safety*, *security* and AVSEC dimensions in isolation, enabling authorities and ANSPs to develop coherent mitigation strategies that avoid organisational silos. Finally, the integration of an artificial-intelligence layer for automatic estimation of the components $(U, V, C)^{f_{\theta}(X)}$ and dynamic adjustment via the anomaly score $V' = V^{(1+\lambda A(X))}$ represents an innovation relative to the static approaches of Elmarady and Rahouma [5], although the lack of operational incident data specific to CNS/ATM systems currently limits empirical validation and parameter calibration, requiring the use of expert estimates or synthetic data.

The conditional-independence assumption between U , V and C adopted in the Bayesian model simplifies modelling but does not yet fully capture the structural correlations observed in CNS/ATM systems, where the most critical elements generally benefit from enhanced protections that mechanically reduce their vulnerability—a limitation that could be addressed in future versions through the adoption of structured Bayesian networks. Overall, the proposed framework offers a relevant compromise between operational simplicity and analytical sophistication, going beyond the limitations of qualitative matrices while remaining more readily deployable than the complex stochastic models mentioned in the literature, and it opens the way to future developments including access to real operational data, integration of hierarchical Bayesian structures and use of advanced AI algorithms to improve the predictive capabilities of the system.

CONCLUSION

This paper proposes an integrated methodological framework for cyber-risk assessment of CNS/ATM systems in civil aviation, based on Bayesian probabilistic modelling of the AVSEC triad threat–vulnerability–consequence. The model combines a quantified risk score $R_{\text{final}} = S + f(U, V, C)$ incorporating operational weights (W_u, W_v, W_c) and a non-linear interaction term $f(U, V, C) = \alpha UV + \beta VC + \gamma UC$ capturing systemic dependencies; a probabilistic classification $P(L \mid U, V, C)$ quantifying the uncertainty associated with each risk level; and an artificial-intelligence layer enabling automatic estimation $(U, V, C)^{f_{\theta}(X)}$ and dynamic adaptation via the anomaly score $V' = V^{(1+\lambda A(X))}$. Simulations on three representative scenarios VHF DoS, GNSS spoofing and CPDLC MITM demonstrate the framework's ability to quantitatively differentiate criticality levels and provide an objective basis for prioritising protection measures, in line with ICAO, EASA and international cyber-risk management standards.

Current limitations include the lack of operational incident data for empirical parameter calibration, the conditional-independence assumption between risk components, and the need for validation on an expanded catalogue of attack sce-

narios. Future research perspectives include building incident datasets specific to CNS/ATM systems, automated integration of threat-intelligence feeds to update Bayesian distributions in near real-time, development of an operational software tool with sensitivity-analysis and scenario-simulation capabilities, extension of the model towards structured Bayesian networks capturing correlations between components, and experimentation with advanced machine-learning techniques to enhance the predictive and adaptive capabilities of the system in controlled operational environments.

REFERENCES

- [1] International Civil Aviation Organization, "Aviation Cybersecurity Strategy," Montréal, QC, Canada, 2019. [Online]. Available: <https://www.icao.int>
- [2] International Civil Aviation Organization, *Air Traffic Management Security Manual (Doc 9985)*. Montréal, QC, Canada: ICAO, 2013.
- [3] International Civil Aviation Organization, "ADS-B Security and Vulnerability Assessment—Technical Note," Montréal, QC, Canada, 2022.
- [4] European Union Aviation Safety Agency, "Cybersecurity in Aviation: Guidelines for an Effective Management of Cybersecurity in the European Aviation System," Cologne, Germany, 2021. [Online]. Available: <https://www.easa.europa.eu>
- [5] A. A. Elmarady and K. Rahouma, "Studying cybersecurity in civil aviation, including developing and applying aviation cybersecurity risk assessment," *IEEE Access*, vol. 9, pp. 143997–144016, 2021, doi: 10.1109/ACCESS.2021.3121230.
- [6] National Institute of Standards and Technology, *Guide for Conducting Risk Assessments*. NIST Special Publication 800-30, Rev. 1. Gaithersburg, MD, USA: NIST, Sept. 2012.
- [7] IEEE Computer Society, *IEEE Standard for Software Life Cycle Processes—Risk Management*. IEEE Std 1540-2001. New York, NY, USA: IEEE, 2001.
- [8] UK Civil Aviation Authority, "Global Navigation Satellite System (GNSS) Outage and Positioning Issues due to Interference," Safety Notice SN-2023/001, London, U.K., Jan. 2023.
- [9] Primature de la République Togolaise, "Arrêté n°2022-040/PMRT du 29 juin 2022 portant adoption des règles de cybersécurité en République togolaise," Lomé, Togo, 2022.
- [10] International Civil Aviation Organization, "The Impact of Artificial Intelligence on the Aviation Sector," Working Paper A42-WP/389, Montréal, QC, Canada, 2025.
- [11] SESAR Joint Undertaking, "AI in ATM: Transparency, Explainability, Conformance, and Performance," White Paper, Brussels, Belgium, 2022.
- [12] International Air Transport Association, "Air Passenger Market Analysis – 2025 Outlook," Montréal, QC, Canada / Geneva, Switzerland, 2025. [Online]. Available: <https://www.iata.org>
- [13] T. Sueki and H. Kim, "Security risk assessment for the Next Generation Air Transportation System," in *Proc. IEEE/AIAA Digital Avionics Systems Conf.*, 2016.
- [14] M. Shaikh, A. A. Rehman, and F. Khan, "Security challenges in enabled aircraft architectures: A survey," in *Proc. IEEE Int. Conf. on Communications (ICC)*, 2019.
- [15] A. Osechas, J. Hernández, and P. Sánchez, "Vulnerabilities of aeronautical radio communication systems: A practical assessment of VDL2 and GBAS," in *Proc. Integrated Communications, Navigation and Surveillance Conf. (ICNS)*, 2017.
- [16] M. Strohmeier, V. Lenders, and I. Martinovic, "On perception and reality in wireless air traffic communication security," *IEEE Transactions on Intelligent Transportation Systems*, vol. 18, no. 6, pp. 1338–1357, 2017.
- [17] M. Strohmeier, V. Lenders, and I. Martinovic, "Securing the air-ground link in aviation," in *Cybersecurity in Aviation*, Springer, 2020, pp. 75–102.
- [18] D. Gritzalis, M. Theodoridou, and G. Stergiopoulos, "Assessing risk in air traffic management as a critical infrastructure," in *Critical Infrastructure Security and Resilience*, Springer, 2019, pp. 145–165.